



Inspiring Excellence

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Department of Mathematics and Natural Sciences
Final Examination
Semester: Spring 2016
Course Title: Integral Calculus and Differential Equations
Course No.: MAT120

Time: 3 hours
Total Marks: 50

Date: April 13, 2016

Note: Answer any THREE from Part A and any TWO from Part B.

Part A: Integral Calculus

1. (a) Evaluate the integral $\int \cos 2x \cos 3x \, dx$. [3]

(b) Sketch the region whose *signed area* is represented by $\int_{-1}^4 |x - 3| \, dx$ and evaluate the integral using an appropriate formula from geometry. [3]

(c) Use x_k^* as the midpoint of each subinterval to find the *area* under the curve [4]

$$f(x) = x^2; -1 \leq x \leq 3.$$

2. (a) Evaluate the integral $\int \sec^4 x \tan^2 x \, dx$. [3]

(b) Evaluate $\int_0^1 \frac{e^{\sqrt{x}}}{\sqrt{x}} \, dx$. [3]

(c) Sketch and find the area of the region enclosed by $y = x^2$, $y = 2 - x$, and $y = 0$. [4]

3. (a) Evaluate $\int \frac{1}{x^2 - 4x + 13} \, dx$. [3]

(b) Find the arc length of the curve given by the following parametric equations. [3]

$$x = e^t(\cos t + \sin t), \quad y = e^t(\cos t - \sin t); \quad 0 \leq t \leq 4$$

(c) Find the volume of solids that results when the region enclosed by the curves $y = x^2$ and $y^2 = x$ is revolved about the y -axis. [4]

4. (a) Evaluate $\int_0^1 \sin^{-1}(x) \, dx$. [3]

(b) Evaluate $\int \frac{2x^2 + 3}{x(x-1)^2} \, dx$. [3]

- (c) Use *cylindrical shells* to find the volume of the solid generated when the region enclosed by the curves $y^2 = x$, $y = 1$, and $x = 0$ is revolved about the x -axis. [4]
5. (a) Define *beta* and *gamma* functions. [3]
- (b) Evaluate $\int_0^\infty e^{-x^2} dx$. [4]
- (c) Find the *area of the surface* generated by revolving the curve given by:

$$y = \sqrt{a^2 - x^2}, \quad a > 0, \quad -a \leq y \leq a$$

about the x -axis.

Part B: Differential Equations

6. (a) How much of carbon-14 (^{14}C) will be left in 100 years if 50 grams are present? Assume that, the half-life of ^{14}C is 5730 years. [5]
- (b) Write the standard form of the first-order linear differential equation. Solve the following differential equation: [5]

$$(x+1)\frac{dy}{dx} + (x+2)y = 2xe^{-x}.$$

7. (a) Use reduction of order to find the general solution of the differential equation [5]

$$y'' + 2y' + y = 0,$$

with one particular solution $y_1(x) = e^{-x}$ and where the prime (') denotes the derivative with respect to x .

- (b) Prove that the following is an exact differential equation. [5]

$$(e^x + y)dx + (2 + x + ye^y)dy = 0.$$

Also solve the equation with the initial value $y(0) = 1$.

8. (a) Solve the following differential equation: [5]

$$y'' + y = \cos x$$

by the method of undetermined coefficients. The prime (') denotes the derivative with respect to x .

- (b) Solve the following initial value problem: [5]

$$y'' + 6y' + 25y = 0, \quad y(0) = 1, \quad y'(0) = 1.$$

The prime (') denotes the derivative with respect to x .



Inspiring Excellence

Department of Mathematics and Natural Sciences
Final Examination
Semester: Summer 2015
Course Title: Integral Calculus and Differential Equations
Course No.: MAT120

Time: 3 hours
Total Marks: 50

Date: August 16, 2015

Note: Question 1 is compulsory. Answer any THREE from Part A and any TWO from Part B.

1. Answer all of the following:

- (a) Is anything wrong with $\int_{-1}^2 \frac{1}{x^2} dx = -\frac{3}{2}$? Explain. [1]
- (b) Are *gamma* and *beta* functions improper integrals? Explain. [1]
- (c) The roots of an auxiliary equation of a homogeneous differential equation with constant coefficients are -2 , -2 , $1+2i$, and $1-2i$. What is the general solution of that differential equation? [1]
- (d) Does $\int_{-1}^2 x^3 dx$ represent the *total* area under the curve $f(x) = x^3$ over $[-1, 2]$? Explain why or why not. [1]
- (e) "If two functions F and G have the same derivative then $F = G$ ". Provide examples to justify or negate the statement. [1]

Part A: Integral Calculus

- 2. (a) Evaluate the integral $\int \sin 5x \cos 7x dx$. [2]
- (b) Sketch the region whose *signed area* is represented by $\int_{-1}^4 2x dx$ and evaluate the integral using the appropriate formula from geometry. [3]
- (c) Use x_k^* as the left end point to find the *area* under the curve $f(x) = 2x - x^2$ over the interval $[-1, 3]$. [4]
- 3. (a) Define definite integral. [2]
- (b) Evaluate $\int_1^e x^2 \ln x dx$. [3]
- (c) Sketch and find the area of the region enclosed by $y = x$, $y = 4x$, and $y = -x + 2$. [4]

4. (a) Evaluate $\int_0^{\infty} \frac{1}{\sqrt{x}(1+x)} dx$. [4]
 (b) Find the arc length of the curve given by the following parametric equations.
 $x = \cos t + t \sin t, y = \sin t - t \cos t; 0 \leq t \leq \pi$ [4]
 (c) Find the volume of solids that results when the region enclosed by the curves $y = x^2$ and $y = 2x$ is revolved about the x axis. [2]
5. (a) State the first fundamental theorem of calculus. [3]
 (b) Evaluate $\int \frac{2x-3}{x^2-3x-10} dx$. [4]
 (c) Use the *cylindrical shells* to find the volume of the solid generated when the region enclosed by the curves $y^2 = x, y = 1$, and $x = 0$ is revolved about the x axis. [2]
6. (a) Define *beta* and *gamma* functions. [3]
 (b) Evaluate $\int_0^1 \frac{x^2}{\sqrt{1-x}} dx$ using beta function. [4]
 (c) Find the area of the surface generated by revolving the curve given by,
 $y = \sqrt{x}, 1 \leq y \leq 4$
 about the y axis.

Part B: Differential Equations

7. (a) Solve the differential equation [4]
 $(e^y + 1)^2 e^x dx + (e^x + 1)^3 e^y dy = 0$.
 (b) Write the standard form of the first order linear differential equation. Solve the following differential equation [5]
 $x \frac{dy}{dx} + (x+1)y = e^{-x} \sin 2x$.
8. (a) Use reduction of order to find the general solution of the differential equation [4]
 $(1-x^2)y'' + 2xy' = 0$,
 with one particular solution $y_1 = 1$ and where prime (') denotes the derivative with respect to x .
 (b) Prove that the following differential equation is *exact* and solve. [5]
 $(\sin y - y \sin x)dx + (\cos x + x \cos y - y)dy = 0, y\left(\frac{\pi}{2}\right) = 0$.
9. (a) Solve the following differential equation by the *method of undetermined coefficients*. [4]
 $y'' + 4y' + 16y = 2x^2 + 3$
 where prime (') denotes the derivative with respect to x and $y_1 = 1$.
 (b) Solve the following differential equation by variation of parameters. [5]
 $y'' + y = \sin x$,
 where prime (') denotes the derivative with respect to x .



MNS Department
Final Examination
Semester: Summer 2014
Course No.: MAT 120
Course Title: Integral Calculus and Differential Equations
(Mathematics II)

Total Marks: 50
Time: 3 hours

Date: August 24, 2014

Section-A: (Integral Calculus)

Answer any **THREE** questions:

1. (a) Estimate the area of the region enclosed by the curve $f(x) = x^2$ within the interval $[0, 3]$ by using the formula:
 $A = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x$ where x_k^* is the midpoint of each subinterval. [5]
- (b) Evaluate the following integrals: [2.5 x 2 = 5]
(i) $\int (\frac{1}{x} + \sec^2 \pi x) dx$, (ii) $\int_0^2 x^2 e^{-x} dx$.
2. (a) Sketch the region whose signed area is represented by $\int_0^4 (12 - 3x) dx$ and evaluate the integral using the appropriate formula from geometry. [5]
- (b) Define convergence and divergence. [1+4 = 5]
Evaluate the integral $\int_0^5 \frac{1}{x-3} dx$ and identify whether it is converging or diverging.
3. (a) Find the area of the region which is bounded by $x = y^2$ and $y = x - 2$. [5]
- (b) Find the arc lengths of the curves given by: [5]
 $x = \cos 2t$, $y = \sin 2t$; $0 \leq t \leq 4\pi$.
4. (a) Evaluate the following integrals: [2.5 x 2 = 5]
(i) $\int \frac{dx}{x^2 + 7x + 12}$, (ii) $\int_0^{\pi/6} \sin^2 6x \cos^4 3x dx$.
- (b) Use cylindrical shells to find the volume of a solid generated when the region in the first quadrant enclosed between $y = x$ and $y = x^2$ is revolved about the y-axis. [5]

[Please turn over]

5. (a) Find the area of the surface that is generated by revolving the portion of the curve $y = x^3$ between $x = 0$ and $x = 1$ about the x -axis. [5]

- (b) Evaluate the following integrals: [2.5 x 2 = 5]

(i) $\int_0^{\infty} \frac{1}{1+x^4} dx$, (ii) $\int_0^b y^5 \sqrt{b^2 - y^2} dy$.

Section-B: (Differential Equations)

Answer any **TWO** questions:

6. (a) Define differential equation. What is the meaning of initial value problem? Solve the initial value problem [1+1+3 = 5]

$$(e^{2y} - y) \cos x \frac{dy}{dx} = e^x \sin 2x, \quad y(0) = 0.$$

- (b) Determine whether the differential equation given by [5]

$$(3x^2 + 4xy)dx + (2x^2 + 2y)dy = 0 \text{ is exact. If it is, solve it.}$$

7. (a) The population of a community is known to increase at a rate proportional to the number of people present at time t . [5]
If the population triples in 8 years, how long will it take to quadruple?

- (b) Use reduction of order to find a second solution $y_2(x)$ of the differential equation $\frac{d^2y}{dx^2} + 9y = 0$; $y_1 = \sin 3x$. [5]

8. (a) Define complimentary function and particular solution. [1+4 = 5]

Solve it: $y'' + y = x \cos x - \cos x$ where $y'' = \frac{d^2y}{dx^2}$.

- (b) Solve it: $y'' + y = \tan x$. [5]

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MNS Department
Final Examinations
Semester: Spring 2014
Course No: MAT 120
Course Title: Integral Calculus and Differential Equations

Marks: 50
Time: 3 hours

Date: 26 April 2014

Section-A (Integral Calculus)

Answer any **THREE** questions:

- Q1.** (a) Use x_k^* as the right end point to find the area under the curve $f(x) = \frac{x^2}{2}$ over the interval $[1, 4]$. 4
- (b) Define definite integral. What is the difference between definite and indefinite integrals? 3
- (c) Sketch the region whose signed area is represented by $\int_{-1}^3 (4 - 5x) dx$ and evaluate the integrals using the appropriate formula from geometry. 3
- Q2.** (a) Sketch the region enclosed by the curves $y = x^2$, $y = \sqrt{x}$ and find its area. 4
- (b) Evaluate the integral $\int_0^b y^5 \sqrt{b^2 - y^2} dy$. 3
- (c) Evaluate the integral $\int e^{2x} \sin x dx$. 3
- Q3.** (a) Find the volume of the solid that results when the region enclosed by the curves $x = \sqrt{1+y}$, $x = 0$, $y = 3$ is revolved about the y -axis. 4
- (b) Evaluate the integral $\int \frac{dx}{x^2 + 3x - 4}$. 3
- (c) Evaluate the integral $\int_0^1 x \sqrt{1-x^2} dx$. 3
- Q4.** (a) Find the arc length of the curve $y = x^{2/3}$ from $x = 1$ to $x = 8$. 4
- (b) Evaluate the integral $\int_0^2 \frac{1}{(x-1)^{2/3}} dx$. 3
- (c) Evaluate the integral $\int_0^\pi \sin^5 \theta \cos^4 \theta d\theta$. 3

- Q5. (a) Find the area of the surface that is generated by revolving the portion of the curve $y = x^3$ between $x = 0$ and $x = 1$ about the x -axis. 5
- (b) Use the cylindrical shells to find the volume of the solid generated when the region enclosed by the curves $y = x^2$, $y = 0$, $x = 1$, $x = 2$ is revolved about the y -axis. 5

Section-B (Differential Equations)

Answer any **TWO** questions:

- Q6. (a) Define differential equation(DE). What is the difference between the solutions of a DE and that of an algebraic equation? 2
- Solve the initial-value problem: $\frac{dy}{dt} + 2y = 1$, $y(0) = \frac{5}{2}$. 2
- (b) Determine whether the differential equation given by $(x - y^3 + y^2 \sin x) dx = (3xy^2 + 2y \cos x) dy$ is exact. If it is, solve it. 3
- (c) Solve the differential equation $(y^2 + yx) dx - x^2 dy = 0$. 3
- Q7. (a) Initially a sample contained 100 milligrams of a radioactive substance. After six hours the mass decreased by 3%. If the rate of decay is proportional to the amount of substance present at time t , find the amount remaining after 24 hours. 4
- (b) Find the general solution of the differential equation $3y'' + 2y' + y = 0$. 3
- (c) Use reduction of order to find a second solution $y_2(x)$ of the differential equation $9y'' - 12y' + 4y = 0$; $y_1 = e^{2x/3}$. 3
- Q8. (a) Solve the differential equation $y'' - 2y' + 5y = e^x \cos 2x$ by the method of undetermined coefficients. 5
- (b) Solve the differential equation $y'' + y = \sin x$ by variation of parameters. 5

THE END



BRAC University
Semester: Summer 2013

Final Examination
Course Title: Integral Calculus & Differential Equations (Mathematics II)
Course No.: MAT 120

Full Marks: 50
Time: 3 Hours

Date: Aug 24, 2013

Section A (Integral Calculus):

Answer any THREE questions:

1. (a) Evaluate (i) $\int x\sqrt{1-x} dx$; (ii) $\int \tan^5 x \sec^4 x dx$. [4]
(b) Find the area under the curve $y = 1 - x^3$ over the interval $[-3, -1]$ with x_k^* as the right end point of each subinterval. [6]
2. (a) Sketch and evaluate the integral $\int_0^1 (x + 2\sqrt{1-x^2}) dx$ using appropriate formula from geometry. [5]
(b) Find the area and sketch the region enclosed by the curve $y^2 = 4x, y = 2x - 4$. [5]
3. (a) Sketch the curve and find the volume of the solid that results when the region enclosed by the curve $y = \sin x, y = \cos x, x = 0, x = \frac{\pi}{4}$ is revolved about the x-axis. [5]
(b) Evaluate $\int_1^3 \sqrt{x} \tan^{-1} \sqrt{x} dx$. [5]
4. (a) Find the arc length of the curve $24xy = y^4 + 48$ over the interval $2 \leq y \leq 4$. [5]
(b) (i) Use cylindrical shell to find the volume of the solid generated when the region enclosed by the curve $y^2 = x, y = 1, x = 0$ is revolved about the x-axis. [2.5]
(ii) Find the area of the surface generated by revolving the curve $x = 9y + 1, 0 \leq y \leq 2$ about the y axis. [2.5]
5. (a) Discuss whether the limit converges or diverges using an appropriate method: $\int_0^{\pi/2} \tan x dx$ [3]
(b) Evaluate $\int \frac{1}{(1-x^2)^{3/2}} dx$ by trigonometric substitution. [3]
(c) Evaluate $\int_0^1 \frac{x^3}{\sqrt{1-x^3}} dx$ in terms of gamma function. [4]

Section B (Differential Equations):

Answer any TWO questions:

6. (a) Find the general solution of the differential equation $x^2y' + x(x+2)y = e^x$. [4]
(b) Determine whether the given differential equation is exact. If it is exact, then solve the initial value problem: [6]
 $(x+y)^2dx + (2xy + x^2 - 1)dy = 0, y(1) = 1$. [5]
7. (a) Solve the differential equation by variation of parameter:
 $y'' + y = \sec\theta \tan\theta$. [5]
(b) Initially 100 mg of a radioactive substance was present. After 6 hours, the mass decreased by 3%. If the rate of decay is proportional to the amount of the substance present at time t , find the amount remaining after 24 hours. [5]
8. (a) Solve the given differential equation by undetermined coefficient: [5]
 $y'' - 10y' + 25y = 30x + 3$.
(b) Solve $xy^2 \frac{dy}{dx} = y^3 - x^3$ while $y(1) = 2$. [5]
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BRAC University
Semester: Fall 2013
Final Examination

Course Title: Integral Calculus & Differential Equations (Mathematics II)
Course No.: MAT 120

Full Marks: 50
Time: 3 Hours

Date: 30th December 2013

Section A (Integral Calculus):

Answer any THREE questions:

1. (a) Sketch and evaluate the integral $\int_{-1}^2 |2x - 3| dx$ using appropriate formula from geometry. [5]
(b) Find the area under the curve $y = 4 - \frac{1}{4}x^2$ over the interval $[0,3]$ with x_k^* as the right end point of each subinterval. [5]
2. (a) Use cylindrical shell to find the volume of the solid generated when the region enclosed by the curve $y = \frac{1}{x}$, $y = 0$, $x = 1$, $x = 3$ is revolved about the y-axis. Sketch the curve. [4]
(b) Find the area and sketch the region enclosed by the curve $x = y^2$, $y = x - 2$. [6]
3. (a) Sketch the curve and find the volume of the solid that results when the region enclosed by the curve $y = \sqrt{\cos x}$, $y = 0$, $x = \frac{\pi}{4}$, $x = \frac{\pi}{2}$ is revolved about the x-axis. [4]
(b) Evaluate $\int_1^3 \sqrt{x} \tan^{-1} \sqrt{x} dx$. [6]
4. (a) Find the arc length of the curve $y = x^{\frac{2}{3}}$ over the interval $1 \leq x \leq 8$. [5]
(b) Evaluate $\int \frac{x^2}{\sqrt{9-x^2}} dx$ by trigonometric substitution. [5]
5. (a) Discuss whether the following integral converges or diverges using an appropriate method: [5]

$$\int_e^{\infty} \frac{1}{x(\ln x)^3} dx$$

- (b) Find the area of the surface generated by revolving the curve $y = \sqrt{x} - \frac{1}{3}x^{\frac{3}{2}}$, $1 \leq x \leq 3$ about the x-axis. [5]

Section B (Differential Equations):

Answer any TWO questions:

1. (a) Separate x and y variables of the differential equation $(e^x + e^{-x}) \frac{dy}{dx} = y^2$. [4]

(b) Determine whether the given differential equation is exact. If it is exact, solve the initial value problem: [6]

$$\left(\frac{1}{1+y^2} + \cos x - 2xy \right) \frac{dy}{dx} = y(y + \sin x), \quad y(0) = 1.$$

2. (a) Solve the initial value problem of the linear differential equation: [6]

$$(x+1) \frac{dy}{dx} + y = \ln x, \quad y(1) = 10$$

(b) Determine whether the given differential equation is homogeneous. If so, solve it by substitution:

$$(x + ye^{\frac{y}{x}}) dx - xe^{\frac{y}{x}} dy = 0. \quad [4]$$

3. (a) Let $y_1(x)$ be a solution of the differential equation given in the following. Find the second solution $y_2(x)$, so that $y_1(x)$ and $y_2(x)$ are linearly independent. [5]

$$y'' + 9y = 0, \quad y_1(x) = \sin 3x$$

(b) Separate t and y variables of the differential equation $\frac{dy}{dt} + 2y = 1$, and solve the initial value problem where $y(0) = \frac{5}{2}$. [5]



BRAC University
Final Examination
Semester: Spring 2013
Course Title: Integral Calculus & Differential Equations (Mathematics II)
Course No.: MAT 120

Full Marks: 50

Date: April 21, 2013
Time: 3 Hours

Section A (Integral Calculus):

Answer any THREE questions:

1. (a) Find the area under the curve $y = \frac{1}{2}x$ over the interval $[1, 4]$ with x_k^* as the right end point of each subinterval. [5]
(b) (i) Evaluate $\int \cos(\ln x) dx$. [2]
(ii) Find the exact arc length of the curve $24xy = y^4 + 48$, over the intervals $y = 2$, $y = 4$. [3]
2. (a) Sketch the region enclosed by the curves and find its area: [5]
 $y = 2 + |x - 1|$, $y = -\frac{1}{5}x + 7$.
(b) Evaluate the integrals:
(i) $\int_1^3 \sqrt{x} \tan^{-1} \sqrt{x} dx$ [3]
(ii) $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \sin \theta \sqrt{1 - 4\cos^2 \theta} d\theta$, by using an appropriate formula from geometry. [2]
3. (a) Find the volume of the solid that results when the region enclosed by the given curves $y = \sqrt{25 - x^2}$, $y = 3$, is revolved about the x -axis. [5]
(b) Evaluate the integrals: [5]
(i) $\int \cos^5 x dx$
(ii) $\int t\sqrt{7t^2 + 12} dt$
4. (a) (i) Use cylindrical shell to find the volume of the solid generated when the region enclosed by the given curves whose equations are given below is revolved about the y -axis: $y = 2x - 1$, $y = -2x + 3$, $x = 2$. [3]
(ii) Evaluate $\int \sin^3 2x \cos^2 2x dx$. [2]

(b) Find the exact area of the surface generated by revolving the curve $y = \sqrt{x} - \frac{1}{3}x^{\frac{3}{2}}$, about x -axis over $1 \leq x \leq 3$. [5]

5. (a) Evaluate the integrals:

(i) $\int \frac{dx}{(4+x^2)^2}$, by trigonometric substitution; [3]

(ii) $\int_e^{+\infty} \frac{1}{x \ln^3 x} dx$, state whether the limit converges or diverges. [2]

(b) (i) Evaluate $\int_0^1 (1 - \frac{1}{x})^{\frac{1}{3}} dx$ in terms of gamma function. [5]

(ii) Evaluate $\int_0^1 x^2 (1 - x^3)^{3/2} dx$ in terms of beta function.

Section B (Differential Equations):

Answer any TWO questions:

6. (a) Find the general solution of $y' + y \tan x = \cos^2 x$. Solve the initial value problem when $y(0) = -1$. [5]

(b) Determine whether the differential equation given below is exact. If it is exact, then solve the following system of equation: [5]

$$\left(\frac{1}{1+y^2} + \cos x - 2xy \right) \frac{dy}{dx} = y(y + \sin x), \quad y(0) = 1.$$

7. (a) Show $x dx + (y - 2x) dy = 0$, is a homogeneous differential equation. Solve the equation by using an appropriate substitution. [5]

(b) Initially 100 ml of radioactive substance was present. After 6 hours the mass has decreased by 3%. If the rate of decay is proportional to the amount of substance present at time t , determine the half-life of the radioactive substance. [5]

8. (a) Separate x and y from $\frac{dy}{dx} = \frac{y^2-1}{x^2-1}$ by substituting the initial values $y(2) = 2$. [5]

(b) Solve the exact equation: $(x - y^3 + y^2 \sin x) dx = (3xy^2 + 2y \cos x) dy$. [5]



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Department of Mathematics and Natural Sciences
Mid-term Examination
Semester: Spring 2016
Course Title: Integral Calculus and Differential Equations
Course No.: MAT120
Sec: 01

Time: 1 hour
Total Marks: 40

Date: Feb 22, 2016

NOTE: Answer any FOUR.

1. (a) Find the following indefinite integrals (any two): [5]
$$\text{i) } \int x^{1/3}(2-x)^2 dx, \quad \text{ii) } \int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx, \quad \text{iii) } \int \frac{1}{4+9x^2} dx.$$

(b) Find the *area of surface* generated by revolving the the curve $y = \sqrt{x}, 1 \leq x \leq 4$ about the x -axis. [5]
2. (a) Use *cylindrical shells* to find the volume of the solid generated when the region R in the first quadrant enclosed between $y = x$ and $y = x^2$ is revolved about the y -axis. [5]
(b) Find the *exact arc length* of the curve $y = x^{2/3}, x \in [0, 8]$. [5]
3. (a) Use the definition of *Area Under the Curve* with x_k^* as the right endpoint of each subinterval to find the area under the curve $f(x) = x^2$ over the interval $[0, 1]$. [5]
(b) Find the volume of solids that results when the region enclosed by the curves $y = x^2$ and $y = 2x$ is revolved about the x -axis. [5]
4. (a) Sketch and find the of area of the region enclosed by the following curves: [5]
$$y = x^2, y = \sqrt{x}, x = \frac{1}{4}, x = 1.$$

(b) Evaluate $\int \sin 3x \cos 4x dx$. [5]
5. (a) Sketch the region whose *signed area* is represented by the definite integral: [5]
$$\int_{-3}^3 \sqrt{9-x^2} dx,$$

and evaluate the integral using *geometry*.
(b) State the *Fundamental Theorem of Calculus (Part I)*. And evaluate any two of the following: [5]
$$\text{i) } \int_0^2 t^2(t-3)^3 dt, \quad \text{ii) } \int_0^{1/\sqrt{2}} \frac{1}{\sqrt{1-x^2}} dx, \quad \text{iii) } \int_1^{\sqrt{2}} \frac{1}{x^2\sqrt{4-x^2}} dx.$$



Inspiring Excellence

Department of Mathematics and Natural Sciences

Mid-term Examination

Semester: Summer 2015

Course Title: Integral Calculus and Differential Equations

Course No.: MAT120

Time: 1 hour

Total Marks: 40

Date: June 26, 2015

Answer any FOUR:

1. (a) Find the following indefinite integrals (any two):

[5]

i) $\int x^{1/3}(2-x)^2 dx$

ii) $\int \sin 3x \cos 4x dx$

iii) $\int \frac{1}{4+9x^2} dx$.

- (b) Find the *area of surface* generated by revolving the the curve $y = \sqrt{x}$, $1 \leq x \leq 4$ about the x -axis.

[5]

2. (a) Use *cylindrical shells* to find the volume of the solid generated when the region R in the first quadrant enclosed between $y = x$ and $y = x^2$ is revolved about the y axis.

[5]

- (b) Find the *exact arc length* of the curve $y = 3x^{3/2} - 1$ over $x \in [0, 1]$.

[5]

3. (a) Use the definition of *Area Under the Curve* with x_k^* as the right endpoint of each subinterval to find the area under the curve $f(x) = 9 - x^2$ over the interval $[0, 3]$.

[5]

- (b) Find the volume of solids that results when the region enclosed by the curves $y = x^2$ and $y = 2x$ is revolved about the x -axis.

[5]

4. (a) Sketch the region enclosed by the following curves and find its area.

[5]

$$y = x^2, y = \sqrt{x}, x = \frac{1}{4}, x = 1$$

- (b) Evaluate $\int e^{2x} \cos x dx$.

[5]

5. (a) Sketch the region whose *signed area* is represented by the definite integral:

[5]

$$\int_{-3}^3 \sqrt{9-x^2} dx$$

and evaluate the integral using *geometry*.

- (b) State the *first fundamental theorem of calculus*. And evaluate any two of the following:

[5]

i) $\int_0^2 t^2(t-3)^3 dt$

ii) $\int_0^{1/\sqrt{2}} \frac{1}{\sqrt{1-x^2}} dx$

iii) $\int_1^{\sqrt{2}} \frac{1}{x^2\sqrt{4-x^2}} dx$.



MNS Department
Midterm Examination
Semester: Spring 2015
Course No: MAT 120 (Section-6)
Course Title: Integral Calculus and Differential Equations

Time: 1 hour

Date: 14 March 2015

Marks: 40

Answer any four from the following questions:

1. (i) Evaluate $\int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$. 5
(ii) Sketch the region enclosed by the curves $y = x^2$, $y = x + 2$ and find its area. 5
 2. (i) Evaluate $\int_2^5 e^{2x-1} dx$. 5
(ii) Use x_k^* as the midpoint of each subinterval to find the area under the curve $f(x) = 4 - x^2$ over the interval $[0, 2]$. 5
 3. (i) Sketch the region whose area is represented by $\int_{x=0}^1 (3x+1) dx$ and evaluate the integral using an appropriate formula from geometry. 5
(ii) Find the volume of the solid generated when the region enclosed by $y = \sqrt{3-x}$, $y = 0$ and $x = -1$ is revolved about the x -axis. 5
 4. (i) Evaluate $\int_{x=0}^1 \sqrt{1+x^2} dx$. 5
(ii) Use cylindrical shells to find the volume of the solid generated when the region enclosed by the curves $y = x^2$, $x = 0$, $x = 2$ is revolved about the y -axis. 5
 5. (i) Find the arc length of the curve $y = x^{3/2}$ from $x = 1$ to $x = 2$. 5
(ii) Evaluate $\int x \cos x dx$. 5
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MNS DEPARTMENT
Midterm Examination
Semester: Summer 2014
Course Title: Integral Calculus and Differential Equations
Course No: MAT 120 (Section-1)

Date: 02 July 2013

Time: 1 hour

Marks: 40

Answer any four of the following questions:

1. (a) Evaluate $\int \sqrt{x} \tan^{-1} \sqrt{x} dx$ using integration by parts. [3.5]
(b) (i) Integrate by trigonometric substitution: $\int \frac{e^x}{\sqrt{1+e^x+e^{2x}}} dx$. [3.5]
(ii) Integrate: $\int \sin^4 x \cos^5 x dx$. [3]
2. (a) Use cylindrical shells to find the volume of a solid that is generated when the region enclosed by the given curves is revolved about the y-axis: $y = 2x - x^2, y = 0$. [5]
(b) Find the area of the surface generated by revolving the given curve about the y-axis:
 $x = 9y + 1, 0 \leq x \leq 2$. [5]
3. (a) Evaluate: (i) $\int [\ln(e^x) + \ln(e^{-x})] dx$, (ii) $\int_0^{\ln 5} e^x (3 - 4e^x) dx$. [3+3.5]
(b) Find the exact arc length of the curve $x = \frac{1}{3}t^3, y = \frac{1}{2}t^2$ over the interval from $t = 0$ to $t = 1$. [3.5]
4. (a) Sketch the region enclosed by the curves $x = y^2, y = x - 2$ and also find the area of the bounded region. [4]
(b) Sketch the function & evaluate the integral using formula from geometry [4]
 $\int_0^3 \sqrt{6x - x^2} dx$.
(c) Indicate u substitution: $\int (3 - \tan x) \sec^2 x dx = - \int u du$ if $u = \underline{\hspace{2cm}}$ and $du = \underline{\hspace{2cm}}$. [2]
5. (a) Use signed area theorem with x_k^* as the left end point of each subinterval to find the area under the curve $f(x) = \frac{x}{2}$ over the interval $[1, 4]$. [5]
(b) State whether the limit converges or diverges: $\int_0^{\pi/6} \frac{\cos x}{\sqrt{1-2 \sin x}}$ [5]

BEST OF LUCK



Department of Mathematics and Natural Sciences
Mid-term Examination
Semester: Fall 2014

Course Title : Integral Calculus and Differential Equations (Math II)

Course No.: MAT120

Section: 09

Time : 1 Hour

Date : Oct 29, 2014

Total marks : 40

Answer any FOUR:

1. Find the following indefinite integrals (any two). [5×2]
i) $\int \tan^{-1} x \, dx$ ii) $\int e^{\sqrt{x}} \, dx$ iii) $\int x^2 \sqrt{1+x} \, dx$
2. a) Use *cylindrical shells* to find the volume of the solid generated when the region R in the first quadrant enclosed between $y = x$ and $y = x^2$ is revolved about the y -axis. [5]
b) Find the *area* of the surface generated by revolving the given curve about the x -axis. [5]
 $y = \sqrt{x}, \quad 4 \leq x \leq 9$
3. a) Find the *volume* of the solid that results when the region enclosed by the given curves is revolved about the y -axis. [5]
 $y^2 = x, \quad y = x - 2$
b) Find the *arc length* of the plane curve over the interval (any one). Leave your answer in exact form. [5]
i) $24xy = y^4 + 48; y \in [2, 4]$ ii) $x = (1+t)^2, y = (1+t)^3; t \in [0, 1]$
4. a) Use the definition of *Area Under the Curve* with x_k^* as the *right* endpoint of each subinterval to find the area under the curve $f(x) = x^2$ over the interval $[0, 3]$. [7]
b) Sketch the region whose *signed* area is represented by the definite integral: [3]
$$\int_{-1}^2 |2x - 3| \, dx$$

and evaluate the integral using an appropriate formula from geometry.
5. a) Sketch the region enclosed by the curves and find its area. [5]
 $y = x^2, \quad y = \sqrt{x}, \quad x = \frac{1}{4}, \quad x = 1$
b) Use *Fundamental Theorem of Calculus* to obtain the following definite integrals (any two). [2.5×2]
i) $\int_1^3 \frac{x+2}{(x^2+4x+7)} \, dx$ ii) $\int_1^e x^2 \ln x \, dx$ iii) $\int_{4\pi^2}^{9\pi^2} \frac{\sin \sqrt{x}}{\sqrt{x}} \, dx$



MNS Department
Midterm Examination
Semester: Spring 2014
Course No: MAT 120 (Sec 04)
Course Title: Integral Calculus and Differential Equations
(Mathematics II)

Time: 1 hour
Total Marks: 20

Date: 12-03-2014

Answer any four of the following questions:

- Q1. Estimate the area of the region enclosed by the curve $f(x) = x^2 + 3$ [5]
within the interval $[0, 3]$ by using the formula:

$$A = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x$$

where x_k^* is the right endpoint of each subinterval.

- Q2. Evaluate the following integrals: [2.5 x2 = 5]

(a) $\int_0^{\pi/8} 2x \cos 2x \, dx$ (b) $\int \frac{1}{(1-3x)^2} \, dx$

- Q3. (a) Find the area of the region which is bounded by $y = 2x + 6$, [2.5]
 $y = x^2$, $x = 0$ and $x = 2$.

- (b) Find the arc length of the curves [2.5]

$$x = \cos 2t, \, y = \sin 2t; \, 0 \leq t \leq 2\pi.$$

- Q4. (a) Find the volume of the solid generated when the region enclosed by the [2.5]
curve $y = x^2$ over the interval $[0, 2]$ is revolved about the line $y = -1$.

- (b) Use cylindrical shells to find the volume of a solid generated when [2.5]
the region in the first quadrant enclosed between $y = x$ and $y = x^2$ is
revolved about the y-axis.

- Q5. Find the area of the surface that is generated by revolving the portion of the [5]
curve $y = x^3$ between $x = 0$ and $x = 1$ about the x-axis.
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MNS Department
Midterm Examination
Semester: Fall 2013

Course Title: Integral Calculus and Differential Equations (Mathematics II)
Course No: MAT 120 (Section-7)

Time: 1 Hour
Marks: 20

Date: 31 October, 2013

Answer any four of the following questions:

1. (a) Evaluate: $\int \sin^3 x \cos^2 x \, dx$. 2
(b) Sketch the region and find the area of the region bounded by the curves $y^2 = 4x$, $y = 2x - 4$. 3
2. Use x_k^* as the **midpoint** of each subinterval to find the area under the curve $f(x) = 4 - \frac{x^2}{4}$ over the interval $[0, 3]$. 5
3. (a) Find the area of the surface that is generated by revolving the curve $y = \sqrt{4 - x^2}$ about the x-axis over the interval $-1 \leq x \leq 1$. 2
(b) Use **cylindrical shells** to find the volume of the solid generated when the region enclosed by the curves $xy = 4$, $x + y = 5$ is revolved about the **x-axis**. 3
4. (a) Sketch the region and use appropriate formula from **geometry** to evaluate the definite integral: $\int_{-3}^0 (2 + \sqrt{9 - x^2}) \, dx$. 2
(b) Find the exact length of the plain curve $y = 3x^{\frac{3}{2}} - 1$ from $x = 0$ to $x = 1$. 3
5. (a) Evaluate $\int \sin^{-1} x \, dx$. 2
(b) Use trigonometric substitution to evaluate: $\int \frac{\sqrt{x^2 - 9}}{x} \, dx$. 3